James Akl

Robotics Researcher

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EDUCATION

Coursework

PhD	Robotics Engineering	Worcester Polytechnic Institute	GPA 4.00/4.00	**Aug 2019**	**May 2024** (exp.)
BEng	Mechanical Engineering	Lebanese American University	GPA 3.72/4.00	**Sep 2015**	**May 2019**
Minor	Mathematics	Lebanese American University	GPA 4.00/4.00	**Sep 2015**	**May 2018**

EXPERIENCE

Doctoral Researcher — Worcester Polytechnic Institute

Manipulation & Environmental Robotics Lab — **Aug 2019 – PRESENT**

Develop strategies for metal scrap cutting, waste sorting, and soft robots using robot **vision**, **control**, and **machine learning**.

Teaching Assistant — Worcester Polytechnic Institute

Robotics Engineering Department — **Aug 2019 – May 2020**

Assisted in the **delivery**, **labs**, and **grading** of these courses: *Unified Robotics I*, *Unified Robotics II*, *Introduction to Robotics*.

Visiting Researcher — University of Ottawa

Mathematical Modeling & Physical Experiments — **Jun 2018 – Aug 2018**

Modeled the vibrations of ribbed plates, and **experimentally verified** the performance of **analytical** versus **numerical models**.

Research Assistant — Lebanese American University

Multiphysics Modeling & Numerical Simulation — **Nov 2017 – Mar 2019**

Demonstrated viability of the underwater ionic buoyancy engine using a coupled multiphysics **numerical model**.

SKILLS

Robotics	Kinematics, Dynamics, Control, Planning, Vision, Learning, Perception, Manipulation, Grasping
Software	C++, Python, Bash, Git, Linux, ROS/Gazebo, MATLAB/Simulink, Mathematica, CMake
Mathematics	Linear algebra, Probability, Statistics, Optimization, Geometric algorithms, Numerical algorithms
Research	Investigation, Simulations, Physical experiments, Surveying/reviewing, Field/on-site deployment
Publishing	MS Office, LaTeX, Technical writing, Diagramming, Video-editing, Presenting
Social	Communication with technical and nontechnical audiences, Cross-disciplinary collaboration
Language	English (native), French (professional fluency), Arabic (professional fluency)

PUBLICATIONS

[2021][3] **ZeroWaste Dataset: Towards Automated Waste Recycling** [Under Review]

[2021][1] **Towards Robotic Metal Scrap Cutting: A Novel Workflow and Pipeline for Cutting Path Generation**

[2021][2] **ECNNs: Ensemble Learning Methods for Improving Planar Grasp Quality Estimation**

[2020][1] **Comparing & Optimizing Analytical, Numerical & Experimental Vibration Models for a Simply-Supported Ribbed Plate**

[2019][4] **Simplified Setup for the Vibration Study of Plates with Simply-supported Boundary Conditions**

[2019][1] **Ionic Buoyancy Engines: Finite Element Modeling & Experimental Validation**

[1–4] Superscripts indicate author position in co-authored papers.

AWARDS

Graduated with Distinction, 2019
LAU School of Engineering

LAU Most Novel Club Award, 2018
Founded the Emerging Technologies Club

LAU Engineering Scholarship, 2015
Won a design and programming competition